
Geography As Exploration and Discovery

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GEOGRAPHY AS EXPLORATION AND DISCOVERY

JAMES J. PARSONS

ABSTRACT. Gold placers attracted initial Spanish settlement to the Antioquia highlands of northwest Colombia. Here there evolved a distinctive culture group, centering on Medellín, that in the nineteenth century fanned out to establish numerous pioneer fronts on Andean slopes. Change has been continuous. Recently Antioqueño colonization has been directed towards the northern tropical lowlands. Medellín, despite locational disadvantages, has become a major industrial center. Through example the author argues for the integrative values of area studies and the significance and rewards of field observation in geography.

WE geographers are an unlikely lot. At times the one common denominator among us seems to be that we call ourselves "geographers," with little consideration as to what that grand word really means. Yet we do have shared values and a common, almost mystic, bond—our curiosity about this planet and the human experience on it—ways of looking at the world that seem to us both unique and worthwhile. Best of all, we are busy doing things that we like to do.

I am among the privileged ones in this fragmented company who find fascination above all in places and people and the interplay between them. In my undergraduate days "geography" never occurred to me, either as a major or a career. As with so many of us, my conversion was due to a chance encounter with a professor who had charisma, wisdom, and a profound "geographical sense." The geography that has most appealed to me has been a kind of his-

torically based "landscape appreciation," and I confess a certain uneasiness with the current compulsion for precision of analysis, the often sterile straining for statistical content and significance.

Geography, so magnificently interdisciplinary, seems an ideal vehicle for the joining of hands of science and humanism, including the taking of moral positions on environmental and spatial issues. This world, after all, is seriously out of balance with regards to production and consumption of food and raw materials, environments are deteriorating, resources and opportunities are unevenly available. As students of the relations between nature and culture we are asking surprisingly few of the really critical questions. How have things gotten this way? Can we possibly use the planet's resources at this rate without using them up, without fouling our nest beyond reclaim? What kind of a jay-cock world is it where annual expenditures for armaments is equal to the entire income of the poorer half of mankind? How long can this and the idiot idea of unending growth, this unquenchable thirst for material goods, be sustained? World population has increased from three to four billion since the A. A. G. met in

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Dallas just fifteen years ago. More people need more food, more energy, more of everything.

Busied elsewhere, we geographers have too largely left to others, and perhaps especially the biologists, the task of exploring the man-nature relationship and facing up to the consequences of the materialistic philosophy with which we have infected so much of the world. Among the most relevant, the most rewarding, of the approaches to our subject, if it is to be a living force, would seem to be the ecological perspective, which is really the long view, from the distant past, into the twentieth century and beyond. It focuses on the earth as the home of man and our responsibilities for the maintenance of its diversity and productivity. We are still an earth science as surely as *geo* is Greek for "earth," yet much of what is being passed off as geography today is simply the study of man, of society, without serious regard for place or time.

THE ATTRACTIONS OF GEOGRAPHY

Many of us are in geography because it involves using our eyes, and for the latitude it allows for wonderment at the world around us, for travel and exploration, for knowing in some depth at least one people and environment other than our own. The excitement of distant places, a colorful map, or the sound of certain place names may first have excited our geographical instincts. In my own case I suspect it was those memorable transcontinental auto trips in high school and college days when considerable stretches of even the Lincoln Highway and Route 66 were still gravel and when gasoline was ten cents a gallon, and even less in East Texas. Or it may have been that stamp collection. I still recall those brightly colored Tanganyikas, with their graceful giraffes browsing on the savanna trees, the caribou on that set from Newfoundland, and our own Columbian Exposition commemoratives of 1892. Whatever the original stimuli, it took. One hopes that at least some of our present recruits are still coming into geography through this door.

Having started out as a newspaperman, I still find myself looking at the world in much the same way that I did as a young reporter on the Mendocino County *Redwood Journal*. The linkage between journalism and geography—at least the geography that interests me, based on observation, conversation and digging through the records leading to a story hope-

fully expressed with clarity and simplicity—is something that is not properly appreciated. Indeed, geography in one sense may be seen as basically a higher form of journalism. Like Bernal Diaz we could well ask if our best hope for immortality might not be that "we wrote the truth and were never dull." We have an unmatched entrée to our supporting public in the naturally given interest of students of all ages in the wonders of the world about us. It would be sad if geography should permit itself to become identified principally as a discipline that can provide techniques and mechanics of control and manipulation for urban, regional, and environmental management. Ours is a major opportunity that transcends mere method. The faculties of description and evaluation are those most in need of cultivation if we are to interpret the relationships of land and life and better illuminate the esthetic qualities of landscapes so that men may live more wisely and happily.

My own focus within geography has come to be the tropical world, especially the American tropics, and, because I live where I do, areas of summer-dry Mediterranean climates. I have come to be impressed, in both life zones, with the significance of vegetation and, in the face of increasing human disturbance, the dynamics of vegetation succession. It is the plant cover, more than anything else, that gives character to the land, to what we see and sense. Vegetation, with its associated wild fauna, is not only a rewarding field for study for its own sake but it also provides an incomparable and sensitive ecologic indicator of the state of the environment, its equilibrium, its productivity, and its potential for supplying human needs, whether for food, industrial raw materials, or simply as wildland to be appreciated and perhaps preserved.

In considering a topic for this occasion, accordingly, I have been strongly drawn to a long held interest, that of the process and consequences of tropical forest destruction, especially the conversion of forest to artificial pasture in Latin America. From here it would have been a short step to the world protein problem, the question of the food producing potential of the earth and especially of the warmer lower latitudes where most of the remaining land reserve lies.

Alternatively, the bicentennial year of the nation seemed to suggest something relating to the American experience, perhaps our west-

ward expansion and its terminal expression out on what Wallace Stegner likes to call "the handsome side of the continent," in the warm folds of California's hills and valleys where America runs out of land. But I was reminded that if 1776 marks the departure of imperialism from the East Coast, it was in the same year that it arrived, under the flag of Spain, on the shores of San Francisco Bay. And that is hardly a theme in the spirit of the Bicentennial!

Instead I have settled on an almost autobiographical review of what I have learned about a particular part of the world with which I have become more or less familiar through recurrent visits and study during the past thirty years. I shall resist cataloguing the intriguing complexity of geographical facts and ecological relationships of the area. Rather, through selected examples, I shall seek to demonstrate what I perceive as the extraordinary advantages of viewing a particular area or set of phenomena over a continuing period of time. It provides the opportunity to observe processes, not at a given moment, but to see them over a substantial period and thus from a variety of perspectives. Even more important to me has been the demonstration of how familiarity and continued exposure may lead to recognition of unexpected themes, problems and relationships within an area—especially an unfamiliar landscape and culture to which one comes with the fresh perspective of an outsider. The significance of what one is seeing may not fully register until it has been repeatedly observed, from different angles and at different points in time. A further and obvious advantage is the growing mastery of sources. Friendships and the confidence of acquaintances mature and yield fruit. With good luck one may come to be considered a part of a community as sympathy and understanding deepen, to be taken into confidence, perhaps encouraged to think that one is doing something that may bring satisfaction and succor to others, not simply exploiting a foreign area as a lucrative research pasture for whatever professional kudos it may promise.

ANTIOQUIA AND THE ANTIOQUEÑOS

I chose northwestern Colombia as my area almost by accident. I had thought, after nearly five years in the military during World War II, that I had earned the right to a year of field study in a region of optimal attractiveness—in my personal ordering of priorities a tropical

highland with shirtsleeve weather, where the mountains were tall and green, the people kindly, and with the colonial Spanish imprint still clearly visible. My original objective had been to study the coffee economy of the mountain province of Antioquia and its impact on the rapidly growing commercial center of Medellín, its orchid-decked mile-high capital (Fig. 1). Vaguely it had occurred to me, too, that its geographical position at the stem end of the southern continent potentially gave it a special significance, for the flood tide of human migrations that swept down from the north to people South America quite clearly funneled through it. But most of all it was an area curiously little known to the outside world, a blank spot on the map of the sort I had always imagined it was up to geographers to explore and to elucidate.

As I came to know the Colombian Andes my concern was diverted from that original focus on the coffee economy, of relatively recent origin, to the phenomenon of the Antioqueños—the people of Antioquia—to the configuration of their culture and the mark it had left on the land.¹ They were derived from a relatively small band of early immigrants, come from Spain in search of gold, who from the first were isolated from the rest of New Granada by barriers of mountains and rain forest and remained so until well into this century. Among other Colombians the Antioqueños stand apart not only for the predominance of Spanish blood (although legend makes them Jewish, or Basque), but for their food, their dress, their language, their politics (Conservative) and their religion (profoundly Catholic). They are a culturally cohesive mountain people living in a very rugged part of the Andes, and their special breed of cattle, their ridge-top rural settlements, their traditional dress, their fondness for tobacco and rum, are all part of the legend and folklore of the "*paisas*," as they call themselves. Their distinctive surnames and the tendency to large and patriarchal families set them strikingly apart from other Colombians even

¹ J. J. Parsons, "Antioqueño Colonization in Western Colombia," *Ibero-Americana*, Vol. 32 (1949); second edition, revised (University of California Press: Berkeley and Los Angeles, 1968); Spanish translation, with prologue and notes, by Emilio Robledo, *La colonización antioqueña en el occidente de Colombia* (Medellín: Imprenta Departamental, 1950); segunda edición (Bogotá: Banco de la República, 1961).

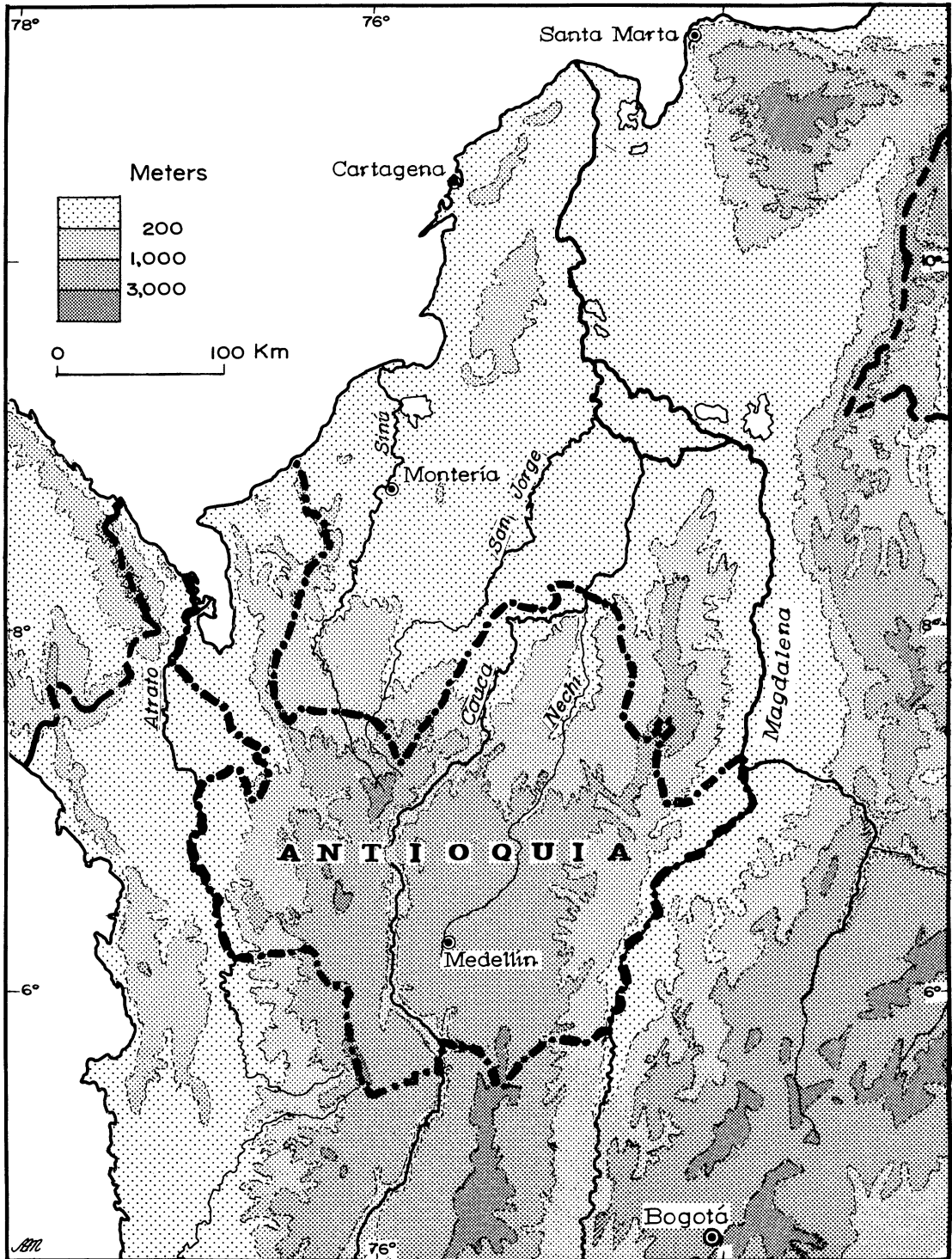


FIG. 1. Northwestern Colombia, with Department of Antioquia.

today. So does their reputation for hard work, frugality, and commercial acumen. These qualities first began to be expressed (after 1790) in the moving frontier of colonization that carried them—with maize-fed hogs and later coffee as the main cash crops—onto the steep, unoccupied volcanic slopes to the south.

An almost Puritanical work ethic over the years was blended with a certain entrepreneurial genius, widely recognized in Colombia, which earned the Antioqueños the nickname "The Yankees of South America." In the nineteenth century it was associated first with gold mining and land clearing and then, in the early twentieth century, with a remarkable surge towards industrialization—against all the odds of geography and economics—which was to make Medellín the first manufacturing city of the Republic and Antioquia its most prosperous department. Most remarkably, it was a process begun when the city's only access to the outside world was still by mule.

The obvious question was why and how this distinctive, energetic, and innovative culture, in many ways so un-Latin, should have evolved when and where it did. After the initial boom in placer gold the Antioquia highlands had gone through two centuries of stagnation. In 1780 they were still being described as the most backward, poverty stricken and isolated part of New Granada, but they were largely malaria-free and racial conflict was at a minimum. There were few Indian survivors of the *Conquista*, and Negro slaves had been confined to the warmer valleys where the placer gold was concentrated. The soils of the granitic batholith of Antioquia were thin and unproductive, like those of New England, and the slopes were inordinately steep, so that there was little interest in accumulating land, and no tendency towards the latifundio so characteristic elsewhere in Latin America. Few large land grants were made or sought. When capital did accumulate there was no thought of putting it into land. Instead, it went into trade and, eventually, into manufacturing industry. Mining, in particular, fostered technological innovation and, with coffee, brought in a certain amount of foreign technology, foreign capital, and, most importantly, led to the establishment of credit institutions, accounting practices, and systems of business organization not previously known to New Granada. Finally, although there were no raw materials to support industrialization, there

was an abundance of hydroelectric power potential and, perhaps, a willingness born of a miner's background for individuals to take the calculated risk. The result was an early and remarkably successful move towards industrialization that has since become a *cause célèbre* among students of economic development seeking to understand how modernization comes to a predominantly peasant society in the face of apparently imposing disadvantages.²

The Antioquia case is of special interest not because it fits a pattern but because it does not. It invokes the very uniqueness that some geographers recently quite purposefully have shunned. Yet to understand better how economic development begins it seems clear that we ought to know more about people who are different in places that are different—such as the Antioqueños of Colombia. The *carteles* that proclaim *Los Antioqueños podemos hacer mas* ("we can do better") reflect a confidence and optimism unusual in Latin America. It is an attitude increasingly seen as worthy of emulation.

Historically, the main frontier of Antioqueño colonization had been southward onto the steep volcanic midslopes of the Andes, into modern Caldas, Tolima and the mountains rimming the

² See L. Ospina Vásquez, *Industria y protección en Colombia* (Medellín: Editorial Santafé, 1955); E. F. Hagen, "The Transition in Colombia," in *On the Theory of Social Change* (Homewood, Illinois: Dorsey Press, 1962), chapter 15; F. Safford, "Significación de los Antioqueños en el desarrollo económico de Colombia," *Anuario Colombiano de Historia Social y de la Cultura* (Bogotá), Vol. 2, No. 3, 1965, pp. 49–70; A. López Toro, *Migración y cambio social en Antioquia durante el siglo XIX* (Bogotá: Universidad de los Andes, Centro de Estudios Sobre Desarrollo Económico, 1968); L. Fajardo, *The Protestant Ethic of the Antioqueños?* (Cali: Ediciones Departamento de Sociología, Universidad del Valle, 1968); W. P. McGreevey, *An Economic History of Colombia, 1845–1930* (Cambridge, England: Cambridge University Press, 1971); F. Chevalier, "Les origines d'un pôle de développement industriel, pour une étude globale du cas de Medellín, Colombia," *Mélanges de la Casa de Velásquez* (Madrid), Vol. 9 (1973), pp. 633–51; A. Twinam, "Antioqueño Entrepreneurship: the Myth and the Reality," *Proceedings from S. U. L. A.* (State University Latin Americanists) (Buffalo: Council on International Studies, State University of New York), Vol. 2 (1973), pp. 184–207. Also, see J. J. Parsons, "Medellín Reconsidered: the Historical Preconditions of Industrialization," *Proceedings of the Conference of Latin Americanist Geographers*, Boca Raton, Florida, December, 1974. Vol. 5 (1976), pp. 119–24.

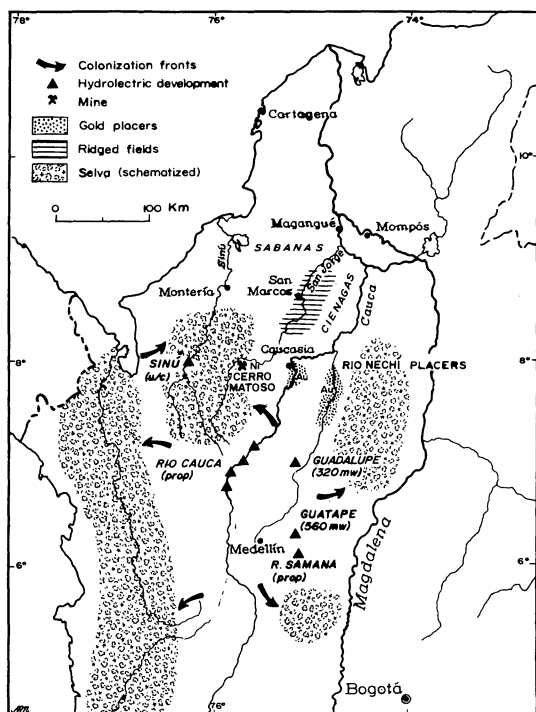


FIG. 2. Colonization fronts, archeologic ridged fields, hydroelectric and mining developments in northwestern Colombia.

Valle del Cauca. In recent decades there has been a new and vigorous thrust northward and northeastward toward the rainy tropical lowlands of the Caribbean coast and the middle Magdalena valley, made habitable by modern medicine and improved transport facilities.³ Here cattle raising, not coffee growing, has been the predominant activity. The clearings in the tropical forest typically yield two or three harvests of yuca, maize, or rice before being converted into planted pasture. Locally in these new lands bananas and cotton are commercial crops. Here in the *tierra caliente* large holdings have been the rule. The Medellín-based capitalist has replaced the small holders, the *minifundistas* even, that characterized the earlier push onto the steep volcanic slopes to the south.

RIDGED FIELDS OF THE SAN JORGE

Observing this later expansion into the hot lowlands I had heard occasional reports of

curious patterns on the land exposed with the clearing of the forest on the floodplains of the lower Río Cauca and especially the Río San Jorge, a lesser Magdalena tributary to the north. Initially it seemed that these lines might be simply the tailings of the California-made gold dredges that work the lower gravels of the streams draining the granitic Antioqueño batholith.⁴ A horseback survey of the area with local *vaqueros* revealed nothing of significance. The fact that William Denevan was turning up evidence of intensive pre-Columbian land use systems on the seasonally inundated lowlands of eastern Bolivia kept the question alive in my mind, but it was not until a chance inspection of air photography of the area between the Cauca and the San Jorge that I became aware that there was indeed something important down there on the ground that no one had ever reported on.⁵ Ground inspection, and reconnaissance by low-flying Cessna, quickly confirmed the existence of more than 70,000 hectares of "ridged fields," systematically laid out by unknown pre-Columbian landscape engineers⁶ (Fig. 2).

As a spin-off from studies of highland colonization we had inadvertently stumbled onto a previously unsuspected form of ancient land use, the intensity of which still puzzles archeologists. It may have significance for the origins of agriculture and the rise of Andean high culture. These raised fields are scarcely visible from the ground or from the dugout canoes in which most travel in the area is performed. Only from the air, and at the time of year when vegetation contrasts are most pronounced, are they clearly observable. The chroniclers were oblivious to them, and even today local resi-

⁴ R. J. Shlemon, *Landslide Terrane near Medellín, Colombia* (Berkeley: Office of Naval Research Report, Department of Geography, University of California, 1970), processed.

⁵ W. D. Denevan, "The Aboriginal Geography of the Llanos de Mojos of Bolivia," *Ibero-Americana*, Vol. 48 (1966); "Additional Comments on the Earthworks of Mojos, Northwest Bolivia," *American Antiquity*, Vol. 28 (1963), pp. 540-45.

⁶ J. J. Parsons and W. Bowen, "Ancient Ridged Fields of the San Jorge River Floodplain, Colombia," *Geographical Review*, Vol. 56 (1966), pp. 317-43; "Los campos de cultivos pre-hispánicos del Bajo San Jorge," *Revista de la Academia Colombiana de Ciencias Exactas, Físicas y Naturales*, Vol. 12 (1966), pp. 449-58. See also L. Deuel, *Flights into Yesterday, the Story of Aerial Archeology* (New York: St. Martin's Press, 1969), pp. 256-62.

³ J. J. Parsons, "The Settlement of the Sinú Valley of Colombia," *Geographical Review*, Vol. 42 (1952), pp. 67-86; "Antioquia's Corridor to the Sea: an Historical Geography of the Settlement of Urabá," *Ibero-Americana*, Vol. 49 (1967).

dents who have recognized them at all are likely to interpret them as natural phenomena. But their linearity and design leaves no room for doubt. They were made by man. The effort they represent raises questions as to their purpose and the kind of strong central authority their construction surely would have required. Although agriculture seems the most obvious explanation, ritual motives cannot be completely ruled out. Their builders shared a mania for earth moving that may relate them in some way to other cultures as far away as Mexico and Bolivia and even the mound builders of the Mississippi valley. Our initial assumption that the ridges were designed to improve drainage in yuca or maize fields may have been too simplistic. In the dry season there may have been sufficient soil moisture in the swales between the ridges to permit practice of a two-crop farming system, the lower and higher surfaces being alternately cultivated. The suggestion that these features may have served as fish traps or fish refuges is untenable. For some it has even been tempting to call on men from outer space. For the moment we are sticking with agriculture, but why such gigantic efforts for such presumably modest returns?

Similar features have since turned up in a number of other areas in the American tropics, in Ecuador, the Venezuelan Llanos, Surinam, the Yucatán peninsula, and even in the high Andean basins of Lake Titicaca and the Sabana de Bogotá.⁷ We are now at work seeking to establish the time parameters for their construction. Recent C-14 dates from organic material in buried soils of the San Jorge ridges suggest that they may have been built in the 300 years immediately prior to the arrival of Europeans (1200–1500 A. D.). But the matter is confused

by a vastly accelerating rate of sedimentation associated with the clearing of the headwater slopes of the Cauca and San Jorge, which appears to have obliterated extensive areas of ridges. At lowest water fossil ridges and swales deeply buried under alluvium have been exposed in profile along the laterally eroding San Jorge and its tributaries. Regional isostatic downwarping in the Mompós Depression, where the Magdalena, the Cauca, and the San Jorge all come together, has been shown to be very rapid, at a rate exceeding one meter in 200 years near the center of the trough.⁸ Who knows what additional expanses of raised fields lie buried there, or how many people and how long it took to build them?

FLOODPLAIN RECLAMATION

Colombian government agencies responsible for land reform and rural development have seen the raised fields as one possible method of intensifying land use on this fertile alluvium. The vast seasonally inundated floodplain of the lower Magdalena valley represents perhaps the largest and best reserve of prime agricultural soils in the country, but periodic flooding has limited their use for the most part to seasonal cattle grazing (Fig. 3). Since the eighteenth century the livestock industry here has been dependent on a classical system of transhumance in which herds spend the dry season in the Ciénagas where water and grass are abundant and are trailed to the higher ground of the Sabanas to the west during the rainy period, when much of the lowland becomes a shallow sea.⁹ During the flood season almost the only dry ground is on the natural levees and an occasional Indian mound or raised field. Most hacenderos in this part of Colombia operate two properties, one in the Ciénagas and one on the higher Sabanas, spending roughly half of the year with their herds on each. Overland movement is difficult during the dry months, impossible during the rainy season. During most of the year the dugout canoe with attached out-

⁷ J. J. Parsons and W. D. Denevan, "Pre-Columbian Ridged Fields," *Scientific American*, Vol. 217 (1967), pp. 92–100; Denevan, op. cit., footnote 5; C. T. Smith, W. D. Denevan and P. Hamilton, "Ancient Ridged Fields in the Region of Lake Titicaca," *Geographical Journal*, Vol. 134 (1968), pp. 353–67; J. J. Parsons, "Ridged Fields in the Río Guayas Valley, Ecuador," *American Antiquity*, Vol. 34 (1969), pp. 256–62; A. H. Siemens and D. F. Poulsen, "Ridged Fields and Associated Features in Southern Campeche: New Perspectives on the Lowland Maya," *American Antiquity*, Vol. 37 (1972), pp. 222–39; A. Zucchi and W. D. Denevan, "Campos agrícolas pre-hispánicos en los Llanos de Barinas," *Indiana* (Berlin), Vol. 2 (1974), pp. 209–25; and B. L. Turner II, "Prehistoric Intensive Agriculture in the Mayan Lowlands," *Science*, Vol. 185 (1974), pp. 718–24.

⁸ República de Colombia, Instituto de Desarrollo de los Recursos Naturales Renovables (INDERENA), "Proyecto Cuenca Magdalena-Cauca: Palinología y Geomorfología," Convenio Colombo-Holandés (Bogotá: INDERENA, 1974), processed.

⁹ H. Wilhelmy, "La cría de ganado en las costas del Caribe del norte de Colombia," *Revista Geográfica*, No. 68 (1968), pp. 63–81; and Parsons, op. cit., footnote 3.

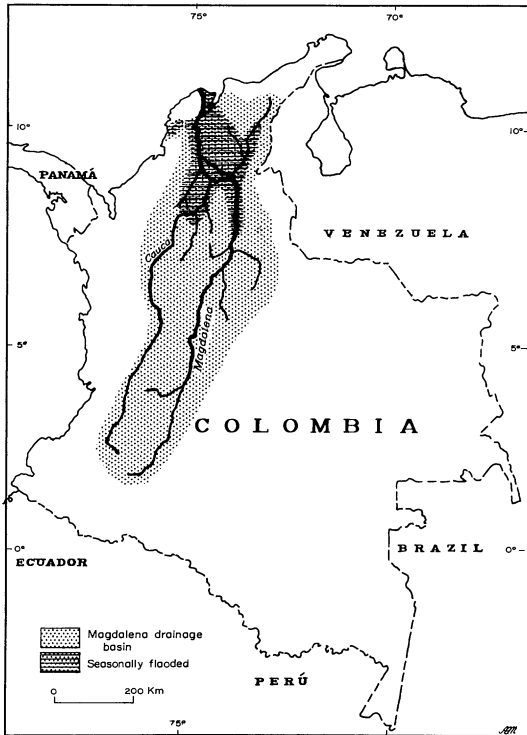


FIG. 3. Magdalena River drainage basin, with seasonally flooded lands.

board, known as "Johnsons" regardless of the mark, replaces the horse and jeep. There is daily launch service between the levee settlements and the Magdalena river entrepot town of Magangué to the north. In another day, when the main course of the Magdalena occupied a now abandoned channel to the east, it was Mompós that played this role. In the colonial era Mompós was one of New Granada's most important cities. Today it lies stranded on a minor arm of the great river without reason for existence, glorying in its illustrious past.

Studies are under way on a master plan to control and use the waters of the Magdalena-Cauca drainage and to reclaim for more productive use the two million hectares in the lower basin that is annually subject to destructive flooding.¹⁰ These fertile alluvial soils near the confluence of the Cauca and Magdalena occupy a privileged geographic position with respect to both markets and transport and they

are already settled with small communities of some campesinos and fishermen. The area possesses at least a minimal economic infrastructure that could be augmented easily.

Between the lower Cauca and the San Jorge alone there are a half million hectares of recent alluvial soils. Forest clearing on the cordillera slopes has brought increasingly destructive floods. In its lower reaches the Cauca today may overtop its banks in any month from May to November. When it does so the floodwaters sweep northward across the low plains toward the San Jorge, funneled by a number of left bank distributary *caños*. Two of these, the navigable Caño Mojana and Caño Panseguita, carry much of this water. The higher ground on their bordering natural levees then offers the only dry ground.

Water, too much or too little, is a matter of life or death in this extensive lowland plain so little known to most Colombians. It is especially during the flood season that tensions mount here. Increasingly, alterations in channel flow and breaks in the Cauca's left bank levee below Caucasia have been intentional. There is a storehouse of legend regarding surreptitious channel diversions and levee dynamiting that reflect long-lived and intensive intercommunity rivalries. A dike breached or a levee thrown up to protect one village may mean flood and devastation to another. Retribution often follows.

The original plan for the reclamation of the Bajo Cauca called for the construction of a 110-km levee along the left bank of the river. This surely would have eventually further aggravated flood problems. More recent studies, by a Dutch-Colombian consortium, indicate such a rapid rate of isostatic subsidence that such large-scale earth moving and channel control is now impractical.¹¹ A series of smaller projects are now being planned in which lesser tracts would be poldered for five to ten years, then replaced by others in a kind of rotating reclamation system. The engineering problems remain formidable. They induce new respect for the ingenuity of those pre-Columbian people who left the raised fields as evidence that man can live in some numbers on this flood-plain environment with a low-energy technology and apparently with minimal ecologic risk.

¹⁰ INDERENA, "Plan de regulación fluvial y defensa contra las inundaciones en la cuenca Magdalena-Cauca," Convenio Colombo-Holandés (Bogotá: INDERENA, 1974), processed.

¹¹ INDERENA, op. cit., footnotes 8 and 10.

AQUATIC AND AVIAN RESOURCES

Some of the earliest agricultural settlements in the New World may have been those along the coasts and rivers of northern Colombia. The evidence for antiquity stems in part from early C-14 dates, and in part on the presence here of an enormously rich aquatic fauna, which would have been exploited easily by primitive man.¹² The system of streams, canals and *ciénagas*, most of which are interconnected, some directly with the sea, is extraordinarily complex. In few places on earth are waterfowl more abundant throughout the year. One species, the blue-winged teal, is a winter migrant from the U. S. and Canada; the others are year-round residents. Manatee, capybara, iguana, agouti, turtle, caiman and *babilla*, as well as abundant freshwater fish, provided a protein base of extraordinary abundance and diversity. Recently these have been under intense commercial exploitation although now U. S. and European prohibitions on imports of their skins are beginning to dry up the export market. The lower Magdalena system is probably the most highly developed inland fishery in South America. The bulk of the catch has been from the great *bocachica* runs every January, February and March, much reduced in late years from overfishing and pollution.¹³ Controls to date have proved ineffective and unenforceable. As a result the fishery appears doomed.

The resident bird population of northwestern Colombia is extraordinarily diverse. Tucked away in the super-rainy Atrato basin, for example, are lowland rain forest species of Amazonian provenience that could only have gotten around the northern spurs of the Andes under much more humid climatic conditions than exist today. With other evidence, as from the pollen record and fossil erosional features, this suggests a substantial shift of climatic belts that may even have had significance for the earliest

human migrations into South America. Both a lower sea level and a northward shift of the equatorial high-rainfall belt are envisioned. Few themes are more significant to the human history of the Americas nor more amenable to biogeographical analysis.¹⁴

FOREST INTO PASTURE

In Colombia as in North America the axe has been the symbol of the pioneer, the forest seen as an enemy to be subdued. Thirty years ago the Antioquia highlands were already largely cleared, except for the areas of shade-grown coffee. Today this is also true of much of the surrounding *tierra caliente*. Soil erosion, including destructive mud flows and landslides, has increasingly muddied the rivers and accelerated deposition rates on the flood plains.¹⁵ Hydraulic mining has further contributed to the desolation. Recently, a massive afforestation program has been undertaken, supported by the World Bank, that is aimed at the establishment of exotic Central American conifers on some 100,000 hectares in the mountains near Medellín. Ten thousand jobs are to be created by the project. The entire appearance of the landscape around the city is thus being drastically altered. "The sense of stewardship for nature," as Archie Carr puts it, "is a mellow virtue that comes long after shoes, and vitamins, and bicycles."¹⁶ It is coming late in Colombia. What makes the afforestation program in highland Antioquia viable is that there is a market for pulpwood and lumber. It will pay.

In the rainy lowlands it is forest removal that pays, most commonly to make way for livestock. Today Colombia's beef cattle population is comparable to its human population—there are about 25 million of each. In the two coastal departments that include the Sinú and San Jorge valleys the ratio of cattle to humans is nearly six-to-one. In the peripheral *tierras calientes* of Antioquia department, in Urabá,

¹² C. O. Sauer early pointed to the significance of riverine habitats and aquatic resources in the establishment of sedentary communities. See *Agricultural Origins and Dispersals* (New York: American Geographical Society, 1952). See also G. Reichel-Dolmatoff, *Colombia* (New York: Praeger, 1965) and review of same by D. Lathrap, "Colombia Prehistory Comes of Age," *Science*, Vol. 152 (1966), pp. 923-25.

¹³ G. Dahl, *Los peces del norte de Colombia* (Bogotá: INDERENA, 1971).

¹⁴ See J. Haffer, "Speciation of Colombian Forest Birds West of the Andes," *American Museum Novitates*, No. 2294 (1967); "Geologic-Climatic History and Zoogeographical Significance of the Urabá Region in Northwestern Colombia," *Caldasia*, Vol. 10 (1970), pp. 603-36.

¹⁵ Shlemon, op. cit., footnote 4.

¹⁶ A. Carr, *Ulendo: Travels of a Naturalist In and Out of Africa* (New York: Knopf, 1964), reference p. 254.

the Bajo Cauca, and the Medio Magdalena, the proportions are similar.¹⁷

Increased livestock numbers have been made possible by the expansion of artificial pasture, almost entirely introduced African grasses.¹⁸ Especially prominent in this have been Guinea grass (*Panicum maximum*), Pará grass (*Brachiaria mutica*), jaraguá or puntero (*Hyparrhenia rufa*), and most recently Pangola (*Digitaria* sp.) and African Star (*Cynodon* sp.). In the Andean foothills molasses grass (*Melinis minutiflora*) and at higher elevations kikuyu (*Pennisetum clandestinum*) have played a similar role. Most of these grasses have been strongly invasive. Their relatively recent introduction to the north coast of Colombia, as elsewhere in tropical America, along with the spread of barbed wire and rotational grazing, has been the basis for a true agricultural revolution that is still in full swing.

Another aspect of the modernization of the north coastal lowlands has been the rapid development of mechanized commercial agriculture. In thirty years the Sinú Valley, long an unknown and mysterious backwater, has become one of the major farming districts of Colombia.¹⁹ It was at one time more familiar to foreigners than to Colombians of the interior. An American company had stripped it of its mahogany and cedar, and there had been an abortive attempt at establishing an export meat-packing industry. Hopes for major petroleum discoveries were never realized, although a small natural gas field near San Marcos supplies the raw stock for the new petrochemical industries of Cartagena. In the end it has been commercial agriculture that has supported the modernization of the Sinú, especially centering on Montería. Cotton is the major crop (the yield from the 16,000 hectares planted in Córdoba in 1975–76 was 36,000 bales), but when the market weakens there is a shift to

rice and sorghum. Pests and floods have been the principal limiting factors in this development. Excessive dependence on cotton pesticides—eighteen applications a year were not unheard of—gradually has been replaced by an integrated pest control program with a consequent sharp reduction in the required number of sprayings. In September, 1975, the worst flood in recorded history devastated the Montería area. There had been others in the past at intervals of twelve or thirteen years. A series of five dams are proposed by the Corporación Electricidad de la Costa (CORELCO) in the upper Sinú that should in part alleviate this problem.

NEW ROUTES TO THE COAST

To Bogotá officials reaching out desperately for ways to close the widening gap between living standard in the cities and the rural areas the integrated river basin planning agency patterned after our T. V. A. has held much attraction. A study prepared under the Canadian-born economist Lauchlin Currie in 1960 led to the establishment of an autonomous regional development corporation for the Magdalena and Sinú valleys.²⁰ The Currie report looked especially to an export-based livestock industry developing on these long neglected lowlands, and the establishment of effective measures of flood control. At the time eighty percent of the area in agricultural use was estimated to be in pasture. The new Ferrocarril del Atlántico, designed to unify the nation's transport net and to provide the first rail link between the Caribbean coast and Bogotá, had been completed five years earlier and major land use changes were clearly in the offing (Fig. 4).

Highway development was to be another important catalyst to economic development of the north coastal lowlands. Thirty years ago the road north from Medellín reached only as far as the Cauca River above modern Caucasia. Beyond, to the north, lay the tropical forest, broken only by patches of edaphic savanna or by occasional man-made clearings. Antioqueño capitalists were already conspicuous in the Montería area, early linked to Medellín by air

¹⁷ S. Kalmanovitz, "El desarrollo de la ganadería en Colombia," *Boletín Mensual de Estadística* (Bogotá), Nos. 253–64, Agosto-Septiembre 1972, pp. 195–241.

¹⁸ J. J. Parsons, "The 'Africanization' of the New World Tropical Grasslands," *Tübingen Geographische Studien*, Vol. 34 (1970), pp. 141–53; "Spread of African Pasture Grasses in the American Tropics," *Journal of Range Management*, Vol. 25 (1972), pp. 13–17.

¹⁹ B. L. Gordon, "The Human Ecology of the Sinú Country of Colombia," *Ibero-Americana*, Vol. 39 (1957); and Parsons, op. cit. (1952), footnote 3.

²⁰ *Programa de desarrollo económico del Valle de Magdalena y norte de Colombia*. Informe de una misión dirigida por Lauchlin Currie según contrato con Robert R. Nathan Associates, Inc., Washington, D. C. (Bogotá: Editorial Argra, 1960).

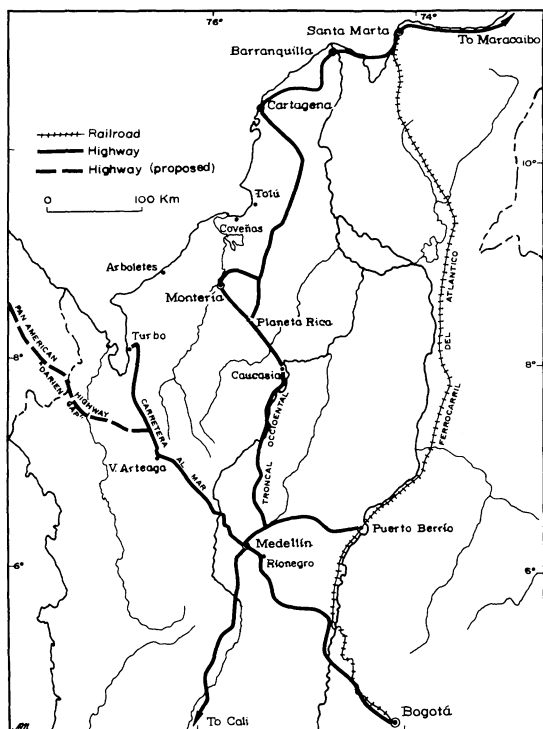


FIG. 4. Major land transport routes, northwest Colombia.

service, and there was agitation for establishment of a new department, to be called Córdoba, after an Antioquia-born patriot, to escape the political domination of Cartagena. At Planeta Rica there was a new airstrip and slaughter house from which daily cargos of chilled beef were being flown to Medellín.

The 680-km Troncal Occidental highway, linking Medellín to the coast at Cartagena, was completed in 1962. Today it is completely paved and handles close to 1,000 trucks daily each way. The population of Antioquia's Bajo Cauca *municipio* of Caucaasia (35,000) has increased ten-fold since the 1951 census.²¹ Montería has exploded from a river village linked with Cartagena only by slow Sinú passenger barges to a provincial capital of 90,000 population (1973 census). The new departments of Córdoba (645,000) and Sucre (189,000) have been carved from the jurisdiction of Carta-

gena.²² Whereas thirty years ago Montería was linked to Medellín only by air cargo and passenger service, today it is a highway hub. The once forested landscape is now all in pasture or field crops. The sandy beaches at the coastal villages of Tolú and Coveñas, along with Cartagena, have become new tourist foci. Antioquia even claims its own Caribbean beach development, at Arboletes, still best reached by small aircraft.

There has been a similar and even more dramatic land opening in the rainy lowlands of the Gulf of Urabá, towards Panamá, where through historical accident the department of Antioquia has a corridor to the sea.²³ Medellín boosters had long seen this as providing a potential outlet for their coffee and surplus manufactures. In 1947, when I first visited the area, the track across the Western Cordillera known as the Carretera al Mar reached only as far as the rubber experiment station at Villa Arteaga, where annual rainfall exceeds 3500 millimeters and there is no real dry season. Turbo was a miserable lumber port on the shallow and muddy waters of the gulf, an emergency stop on the pioneer UMCA air line from Panamá to Medellín. It was in 1954 that the first vehicles arrived on the Urabá shores over the tortuous 350-kilometer mountain road, but numerous uncompleted bridges continued to present problems. Six years later the United Fruit Company, discouraged by blowdowns and labor unrest at Santa Marta, came into the Urabá area with a radically new concept of banana growing. Instead of producing fruit on its own plantations, the company here became contractor and buyer, granting easy credit to investors from the interior who were in turn guaranteed a market for their fruit. Shipments began in 1964. Access roads, drainage canals, technical advice, and helicopters for spraying against Sigatoka disease, even the Gros Michel seed clones for planting, were provided by the company. The population of the Urabá *municipios* of Antioquia rose from 14,435 in 1951 to 126,406 in the 1973 census and it is still growing rapidly. It is one of the four or five leading

²¹ E. Gühl, "Anotaciones sobre población, poblamiento, posición y estructura demográfica en Colombia," *Revista de Academia Colombiana de Ciencias Exactas, Físicas y Naturales*, Vol. 12 (1966), pp. 377-90.

²² S. García, *Geografía del Departamento de Córdoba* (Medellín: Bedout, 1975); and *Monografía del Departamento de Sucre* (Bogotá: Instituto Geográfico 'Agustín Codazzi,' 1969).

²³ Parsons, op. cit. (1967), footnote 3.

banana producing districts of the world. Its "associate producer" program has become a model for the fruit company, now United Brands, in other producing areas. A 3,000-hectare planting of African oil palms at the Dutch-managed Coldesa estate near Turbo, originally decimated by bud rot, has been replanted with a disease resistant interspecific hybrid between the African oil palm (*Elaeis* spp.) and the closely related *Corozo oleifera*, a native of the moist tropics of South and Central America, apparently with promising results. Marihuana, cultivated clandestinely for smuggling into the United States, has been an important source of cash for settlers in the more remote hill districts, where land clearing is still going on. Urabá has all the problems of a tropical frontier—land concentration, inadequate infrastructure, uncertainty of land titles, and the rest—but after ten years the banana economy is still alive and well and boosters still envision a major Caribbean port here, Antioquia's own outlet to the Atlantic world.²⁴

The new highways from Medellín to the sea, and even the *autopista* being pushed southeastward towards the Magdalena river and Bogotá, have a certain parallel with the penetration roads that elsewhere lead down from the Andes into the eastern tropical lowlands.²⁵ These have been major generators of migration and lowland settlement but the routes out of the Antioquia highlands serve as outlets, as windows on the Atlantic world, instead of terminating as do the others in cul de sacs within the Amazon rainforest. They are not exactly freeways (maintenance is a major problem) but they are two-way routes supporting a considerable two-way commerce. As a result the development of the areas they transect is dependent as much on advantageous location as on the productivity of their soils. The clearing and settlement process has been much faster, the elimination of malaria more decisive, and the establishment of a viable infrastructure easier than in the *tierra caliente* colonization areas east of the Andes.

²⁴ S. Z. Goldhaber, "Medicine without Frills: a Rural Hospital in Colombia," *Science*, Vol. 189 (1975), pp. 1067–71, provides a notable account of life and death in the Urabá 'boom town' of Apartadó.

²⁵ R. C. Crist and C. M. Nissly, *East from the Andes: Pioneer Settlements in the South American Heartland* (Gainesville: University of Florida, Social Science Research Monograph 50, 1973).

THE "DARIEN GAP"

The almost trackless tropical forest of the southern extremity of the Panamá isthmus, with the adjacent Atrato swamps of Colombia, is the last remaining break in the Pan-American Highway system. This is true despite the proximity of the formidable technological might of the U. S.-Panama canal organization and the extensive recent investigations for a possible second transisthmian canal, either in southeast Panamá or in Colombia. A 90-km ferry service between Colón and Turbo has been proposed by Medellín interests to break the "Darien Gap," but diplomatic barriers have arisen. Meanwhile highway construction lags. U. S. financing, which has been approved, carries the requirement of a detailed environmental impact statement. There is concern that Central America will be opened wide to hoof-and-mouth disease and other pathogens should a land route be opened, and that destruction of a near virgin tropical forest ecosystem and its sparse Indian populations will inevitably ensue.²⁶ A shorter, easier land route along the San Blas coast, with a ferry across the Gulf of Urabá to Turbo, has been urged as a compromise but it has been blocked by politically powerful landowners in Panamá and by bureaucratic inertia. When the Pan-American Highway is completed, as eventually it will be, the anticipated traffic to the South American continent will all funnel through northwest Colombia. A proposed coastal highway (Ramal del Caribe) linking up with Cartagena, Barranquilla, Maracaibo, and Caracas, will probably supplement the present tortuous route over the mountains to Medellín, thus introducing the prospect of yet another revolutionary geographical reorientation of the regional economy.

ELECTRICITY AND GOLD

The early industrial development of Medellín, largely based on textiles, was facilitated by the availability of water power. Energy con-

²⁶ A. R. Hanbury-Tenison and P. J. K. Burton, "Should the Darien Gap be Closed?," *Geographical Journal*, Vol. 139 (1973), pp. 43–52; R. Soles, "Pan-American Highway: Running a Dream Through a Swamp," *Nation*, February 21, 1972, pp. 233–36; and "Final Environmental Impact Assessment, Darien Gap Highway from Tocumen, Panamá, to Río León, Colombia," (Washington, D. C.: Federal Highway Administration, U. S. Department of Transportation, 1974).

tinues to be abundant and cheap. Medellín consumes daily half again as much electricity as the much larger Bogotá. The Empresas Públicas de Medellín has developed major hydro-electric complexes both east and west of the metropolitan area, at Guadalupe and Guatapé, that have a combined installed capacity of nearly one million kilowatts. A further one and one-half million kilowatts is projected in the Río Samaná scheme. Still other undeveloped sites remain in these rainy highlands, especially the deep, strike-oriented canyon of the Río Cauca as it cuts across the northernmost reaches of the Antioquia batholith. Here preliminary plans call for six major dams with an eventual additional generating capacity of an astonishing ten million kilowatts, the equivalent of five Grand Coulees. On the upper Sinú the complex of dams planned, when completed, will add another one and one-half million kilowatts, much of which is earmarked for a major nickel mining and smelting venture at Cerro Matoso in the upper San Jorge drainage where Hanna Mining Co., in association with the Colombian government, is opening up a sixty-five million ton nickel laterite ore reserve. Iron ore and cobalt will be by-products. Start up of the estimated \$88 million venture is currently envisioned for 1978, probably with Japanese participation. It promises further major economic reverberations throughout the area.

Gold mining led to the initial settlement of Antioquia and, along with the looting of Indian graves (*guaquería*), long supported its economy. The high bench gravels and the vein deposits of the granitic batholith were augmented by the auriferous outwash gravels in the *tierra caliente*, especially on the lower Cauca and Nechí rivers in the vicinity of Cáceres and Zaragoza.²⁷ During the present century these rich gravels have been hydraulicked and dredged by American interests, especially on the lower Nechí, where five giant dredges still operate, now under Colombian ownership.²⁸ The recent sharp rise in the price of gold has given new life to the entire area. Small-scale hydraulic and sluicing operations have proliferated, and hand panning with wooden *bateas*

has returned. In 1975 encampments of some 5,000 *barequeros* were placing in one pasture in the vicinity of Caucaasia. They were recovering \$2–5 (U.S.) a day, substantially better than the going ranch wage. With the price of gold what it has been, panning or sluicing productive auriferous gravels turns out to be a very attractive proposition in a peasant economy. It's the Gold Rush in the Mother Lode of California all over again.

ISLANDS AND SEA TURTLES

And so, from our original and tentative concern for the coffee economy of the Colombian Andes I have found myself diverted largely by chance and coincidence to a considerable range of geographic themes. There is one further link to this personal chain of associations. In the course of field study in a remote corner of the Sinú valley I had come upon several black Colombians of distinctive demeanor who spoke a kind of soft creole English. In Cartagena there were others of their like, bilingual cab drivers and tour guides. They came, I soon learned, from the Colombian Caribbean islands of San Andrés and Providencia, and I was attracted by the invitation to visit these English-speaking Protestants in their remote insular homes off the Nicaragua coast, then accessible only by sailing vessel from Colón or Cartagena. My study of the islands was fortuitously completed just prior to the dramatic transformation of their economy and culture brought by the arrival of jet air service and freeport status.²⁹ It has served as a baseline for several later studies and may have heightened the awareness of Colombians of the existence of a seldom recognized form of internal "imperialism."

This pleasant foray into the historical geography and cultural particularism of this distinctive Caribbean island world was not the end. It was here, and more precisely on the beach at Providencia, that I first learned about sea turtles, especially the giant green turtle of

²⁷ Parsons, op. cit. (1949), footnote 1; and R. C. West, *Colonial Placer Mining in Colombia* (Baton Rouge: Louisiana State University Press, 1952).

²⁸ R. J. Shlemon and B. Phelps, "Dredge-tailing Agriculture on the Río Nechí, Colombia," *Geographical Review*, Vol. 61 (1971), pp. 396–414.

²⁹ J. J. Parsons, "San Andrés and Providencia: English-speaking Islands in the Western Caribbean," *University of California Publications in Geography*, Vol. 12 (1956), pp. 1–84; Spanish translation, with notes, by F. Archbold Britton, *San Andrés y providencia: una geografía histórica de las islas colombianas del mar Caribe occidental* (Bogotá: Banco de la República, 1964). Also, J. J. Parsons, "English-speaking Settlements of the Western Caribbean," *Yearbook, Association of Pacific Coast Geographers*, Vol. 16 (1954), pp. 3–16.

turtle soup fame and the hawksbill that provides the tortoiseshell of commerce.³⁰ Continuing inquiry into the exploitation of these remarkable reptiles, so important in the early history of the Caribbean, led to consideration of them in the world context. Once plentiful, they are now an endangered species as the result of increasing pressures being relentlessly imposed on them, especially at their highly localized nesting beaches. Today I am afraid that some of my friends think of me principally in terms of turtles. Perhaps that is the inevitable fate of an eclectic geographer with a predilection for the tropics.

CONTEMPORARY ANTIOQUIA

To return to the Antioqueños, their department, so long the bellwether of the Colombian economy, is today losing ground.³¹ In manufacturing it has been overtaken by Bogotá, and Cali is seriously challenging it for second place. Economic stagnation, social disintegration, and environmental deterioration are a part of the *realidad Antioqueño* of the mid 1970s as the population of the Valle de Medellín pushes one and one-half million. The gross disparities of wealth, income, and influence between the metropole and the impoverished rural *municipios* continues deeply troublesome. A remarkable process of self-analysis and self-criticism organized by private civic, church, and industrial groups is under way.³² The problems being addressed include the future of manufacturing as the driving motor of the departmental economy, the lagging traditional agricultural sector of the highlands and its excessive dependence on coffee, the harnessing of the new dynamism of the

peripheral lowlands, and the continuing concentration of people and activities in the topographically constricted valley occupied by the Medellín metropolitan agglomeration which threatens seriously the quality of life. There have been dramatic increases in rural violence and kidnapping and continued growth of the economically and socially marginal sector of the urban population. The establishment of a major new satellite city, either at Rionegro on the high Oriente plateau, to be linked with Medellín by a six-kilometer inclined tunnel, or in the eastern lowlands along the Río Magdalena, has been proposed.³³ Only the continuing, relentless departmental population growth, almost wholly attributable to the continuing high rate of natural increase, has not been squarely faced.

For the first time there is recognition that Antioquia's isolation from the rest of the country, in psychological and socioeconomic terms as well as physically, may be taking a toll. The traditional closed society of Medellín, with its air of self-complacency and its persistent demands for regional autonomy, has won the Antioqueños few friends. A mentality largely closed to outside influences, whether from overseas or from other parts of the country, especially the capital, has become increasingly counter-productive in a world of expanding national and international relationships and markets. The issues are being boldly confronted as the well-known creativity and organizational genius of the Antioqueños is challenged to respond to an unexpected and unfamiliar set of circumstances. The cry for an "independent Antioquia" continues to be heard but it has relatively few followers.³⁴

ON GOING BACK

In my corner of South America, and I suspect elsewhere, everything in the end has seemed to be hitched to everything else. I have found that it pays to keep going back to an area, a people. I have found that significant phenomena or relationships continue to present themselves, things that were at first completely missed or whose significance was not originally apparent. In my experience these linkages are

³⁰ J. J. Parsons, *The Green Turtle and Man* (Gainesville: University of Florida Press, 1962); and "The Hawksbill Turtle and the Tortoise Shell Trade," *Études de Géographie Tropical Offertes à Pierre Gourou*, (Paris, La Haye: Mouton, 1972), pp. 45-60.

³¹ J. Restrepo Uribe, *Antioquia: Olvidada, marginada y resentida* (Medellín: Bedout, 1964); A. Gilbert, "A Note on the Incidence of Development in the Vicinity of a Growth Centre," *Proceedings of the Conference of Latin Americanist Geographers*, Boca Raton, Florida, December, 1974, pp. 35-50; G. Ibiza de Restrepo, *El proceso de desarrollo económico de Antioquia* (Medellín: Instituto de Integración Cultural, 1975).

³² *Encuentro de Quirama* (Medellín: Acción para el desarrollo de Antioquia, 1974); *Antioquia en Urabá* (Medellín: Acción para el desarrollo de Antioquia, 1974).

³³ "El tunel de Rionegro, obra nacional," *El Colombiano*, Medellín, 21 Mayo, 1975.

³⁴ J. Restrepo Uribe, *La República de Antioquia* (Medellín: Bedout, 1972).

quite accidentally discovered. They are seldom of the sort that can be conjured up in one's study. Repeated visits and growing familiarity with an area continually enlarge one's range of vision and concern. You see, you hear, or read something new, something out of place, something irregular, and you are moved to investigate further. There may be many false leads, many a dead end, but if one is lucky and persists there may be undreamed of rewards awaiting. The experiences, the acquaintances made along the way, can make geography enormously good fun. You learn the ropes, how to get around in the libraries and archives, how to use the local newspapers as the rich sources of information they often are, who the people are who know their area, who are at heart the "geographers" with a reservoir of local knowledge and insight above and beyond the norm. In such company time moves fast. Life becomes worth living. As Carlos Castañeda's "Don Juan" says "We are going to be here in this stupendous, awesome, unfathomable world only a short while, too short for witnessing all the marvels of it." We have got to learn to make every act count, even though it may on occasion mean taking risks, living dangerously.

AREA AS INTEGRATING CONCEPT

For me, at least, it all comes back to place, to area as the integrating concept, to the supremacy of observed geographic data over any pyramid of deductions or formal theories, however powerful the apparatus brought to bear. Geography by its nature offers the promise of carrying us beyond generalizations to more exact knowledge of the interaction of man and environment. Diversity, we are learning, seems to be a major stabilizing force in the natural order, and probably in the social order, too. It is something to analyze and understand, not something to be gotten rid of. Interest in the unique, moreover, may be the single most powerful recruiting agent we have for bringing students into geography. We must be aware of those who would brainwash us into thinking it is something shameful or secondary, that only generalizations or theory are wholly respectable.

It does of course take time, access to a minimum of travel, money (and here a personal bow to my own university and to the Office of Naval Research for support), and perhaps a certain perverse propensity for roughing it.

Having one's share of good fortune can help, too, perhaps beginning with choosing the right area in which to work. Rosalie Wax, in an autobiographical review of her own ethnographic field career, observes that the "most valuable thing any fieldworker can take with him into the field is good luck."³⁵ Luck, she suggests, is a gift of the Powers and cannot be acquired by determination or study. Being at the right place at the right time is part of it, but it also helps to know that you are there. The next most useful thing to take into the field, she continues, is intelligence manifested as common sense, shrewdness, and flexibility—the property called "having one's wits about one." "I cannot tell a student how to acquire or develop wits," she writes, "but I can tell him this: that if he does not have his wits about him while in the field the chances are that he will not be given the opportunity to exercise any of his other attributes or virtues."

As we seek out paths to expanded awareness through teaching or writing we would do well not to forget that our subject was originally rooted in the comparative observation and analysis of places. The field experience that such geography almost inevitably entails has the special virtue of lifting us quite decisively from the quagmire of definitions of our subject and from methodology and it gets us back into the open air. It reaffirms the spirit of geographical adventure and the validity of personal observation and of intuitive knowledge. It may encourage the integration of large general themes, the weaving together of disparate strands into a tapestry of land and life in a regional context that is both satisfying and useful.³⁶ It makes us, I suppose, as much humanists and historians as scientists. But it may just be that such knowl-

³⁵ R. H. Wax, *Doing Fieldwork: Warnings and Advice* (Chicago: University of Chicago Press, 1971), reference p. 268 (cited in J. C. Friberg, "Field Techniques and the Training of the American Geographer," Discussion Paper Series No. 5, Department of Geography, Syracuse University, 1975, p. 15).

³⁶ The points of view here expressed have found repeated affirmation in the works of C. O. Sauer. See also, D. W. Meinig, "Environmental Appreciation: Localities as a Humane Art," *Western Humanities Review*, Vol. 25 (1971), pp. 1–11; and D. Hooson, "Rejuvenating Regional Geography: Ends and Means," *Proceedings, 21st International Geographical Congress*, New Delhi. Vol. 4 (Calcutta, 1972), pp. 91–94.

edge and the understanding deriving from it is what the future will most want and expect from us. At least I detect a current flowing that way. I hope that I am right, that geographers to come may continue to experience the joys of exploration and discovery that have been one of the wellsprings of our subject's strength since Antiquity.